Introduction to Deep Learning and Foundation Models for Time Series: Towards Better Generalization at Inference

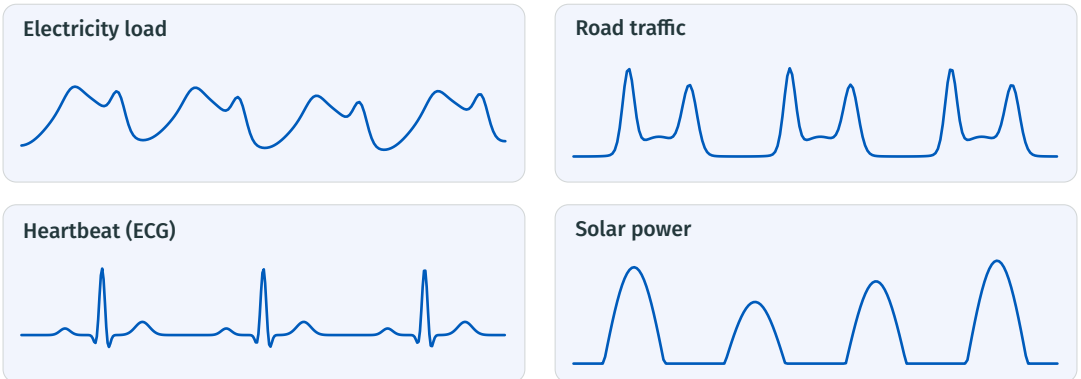
AI for Industry Workshop

4 November 2026

Etienne Le Naour	EDF R&D
Tahar Nabil	EDF R&D
Adrien Petralia	Mistral AI

Introduction

Time series are ubiquitous



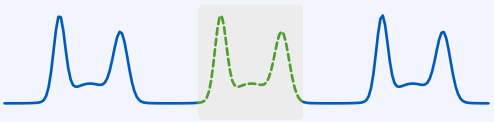
A time series: $\mathbf{X} = (\mathbf{x}_1, \dots, \mathbf{x}_T) \in \mathbb{R}^{C \times T}$, C channels, T time steps

One object, many tasks

Forecasting



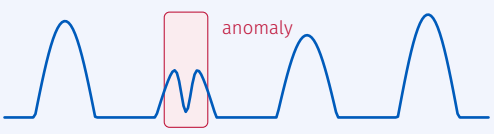
Imputation



Classification / regression



Anomaly detection



...and many others: **segmentation**, **disaggregation**, **generation**, clustering, change-point detection, ...

In today's talks: four tasks

Forecasting



predict what comes next

Imputation



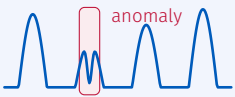
fill in missing values

Classification



put a label on a whole series

Anomaly detection



spot unusual behaviour

In today's talks: four tasks

Forecasting



predict what comes next

Imputation



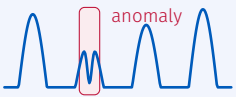
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Classification



put a label on a whole series

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spot unusual behaviour

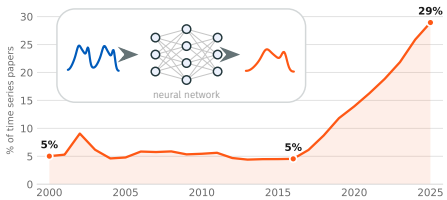
Statistics 1950s–
exponential smoothing,
ARIMA, Kalman filters

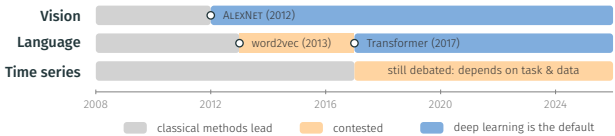
Machine learning 1990s–
nearest neighbours + DTW,
random forests, boosted trees

Deep learning 2010s–
RNNs, CNNs, Transformers,
diffusion models

Decades of research on each of them: deep learning is only the **latest wave**

Deep learning takes over time series research... but more slowly than in language





Share of time series papers using neural networks. Source: OpenAlex.

It depends on the task

generation imputation

deep learning leads

forecasting classification

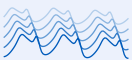
still contested

It depends on the data



few, short series

statistics



many related series

deep learning

Evaluation practices

"Not (yet) as good as hoped"

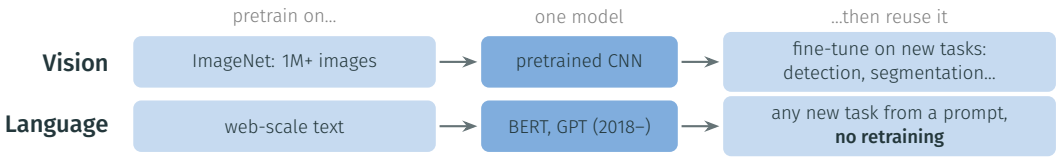
Bergmeir [2024]

Weak baselines, **flawed evaluation**

[Hewamalage et al., 2023]

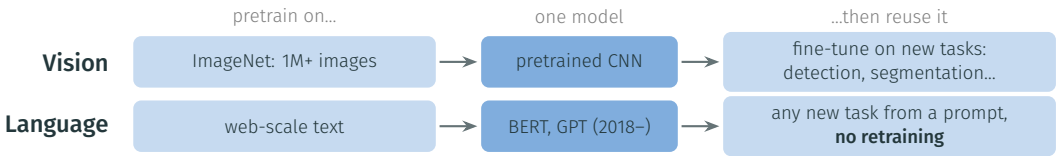
What made deep learning the default elsewhere: large-scale pretraining

Deep learning needs data... but most time series use cases come with **few, short series**



What made deep learning the default elsewhere: large-scale pretraining

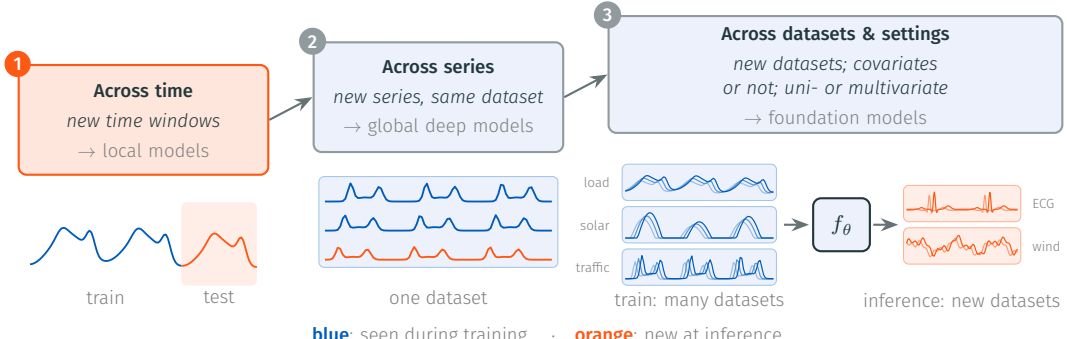
Deep learning needs data... but most time series use cases come with **few, short series**



Pretrain on lots of data, then **reuse** the model: does this work for time series?

Roadmap: three levels of generalization

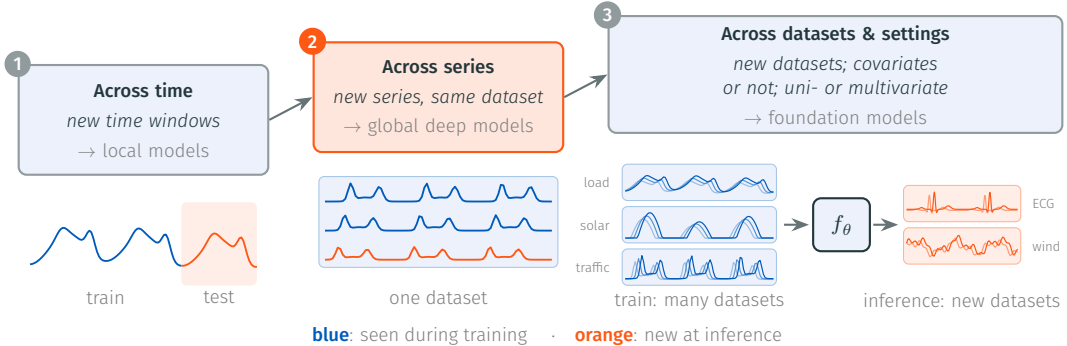
Generalize = how far we can go from the training data **without retraining**



Level 1, **statistics**: one model per series, fitted on its own past

Roadmap: three levels of generalization

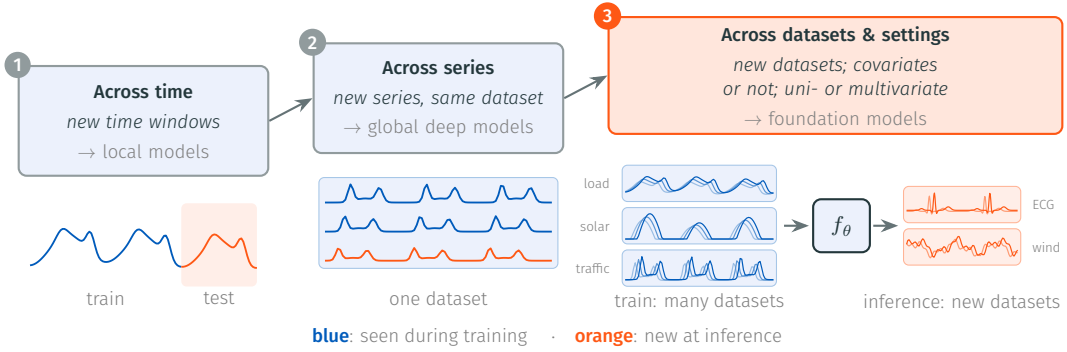
Generalize = how far we can go from the training data **without retraining**



Part I: traditional deep learning, **one model per dataset**
Part II: why it does not scale

Roadmap: three levels of generalization

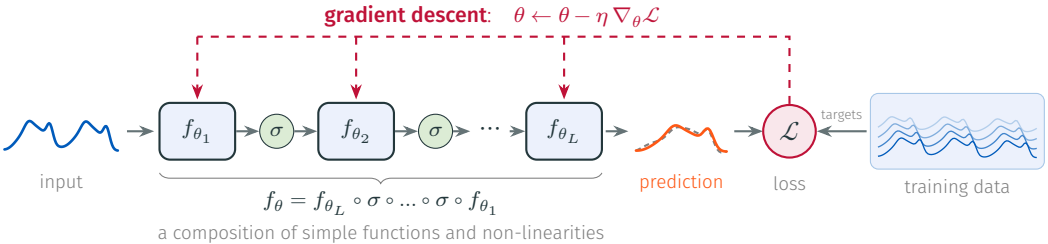
Generalize = how far we can go from the training data **without retraining**



Part III: foundation models, **new datasets without retraining**

Part I: Traditional Deep learning for Time Series

Deep learning in a nutshell



Learnable compositions

Layers + **non-linearities**,
all trained by **gradient descent**

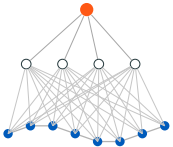
Only as good as the data

The data sets the ceiling:
quality, **quantity**, **diversity**

Various mechanisms and architectures

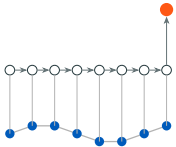
Mechanisms

how information flows along time



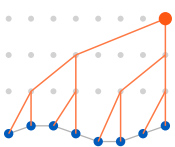
MLP

sees the whole window



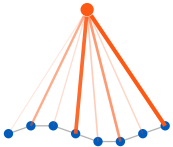
RNN

state updated step by step



CNN

local, shared filters



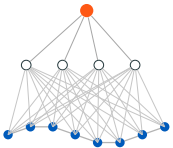
Attention

weights on every step

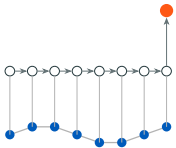
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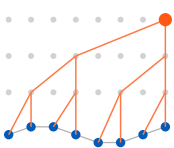
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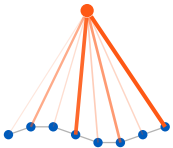
MLP
sees the whole window



RNN
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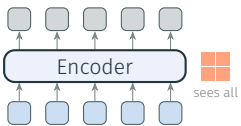
CNN
local, shared filters



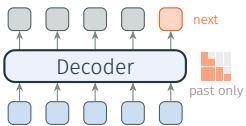
Attention
weights on every step

Architectures

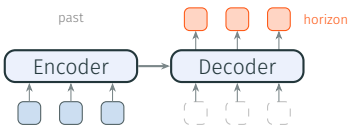
how the blocks are assembled



Encoder-only



Decoder-only



Encoder-decoder

Zoom on forecasting (1/2): attention between single time steps

Language

TRANSFORMER (2017)



a token = a **word**: it carries meaning on its own

Zoom on forecasting (1/2): attention between single time steps

Language

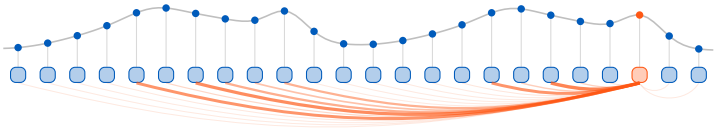
TRANSFORMER (2017)



a token = a **word**: it carries meaning on its own

Time series

INFORMER, AUTOFORMER, FEDFORMER (2021–22)



a token = **one time step**: a single value, little meaning on its own

·INFORMER: sparse attention · AUTOFORMER: auto-correlation · FEDFORMER: attention on Fourier modes

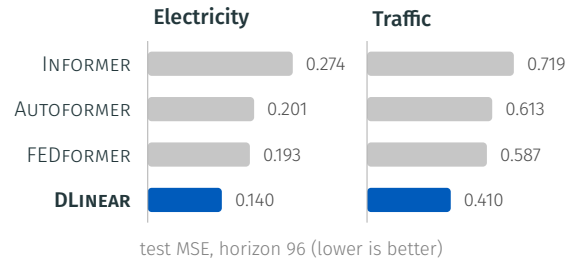
one token per time step

[Vaswani et al., 2017, Wu et al., 2021, Zhou et al., 2021, 2022]

Same recipe as language, **one token = one time step**... but a single value means **far less than a word**

Zoom on forecasting (2/2): “Are Transformers effective for TS forecasting ?”

One **linear layer** beats them all



Zeng et al. [2023], Table 2

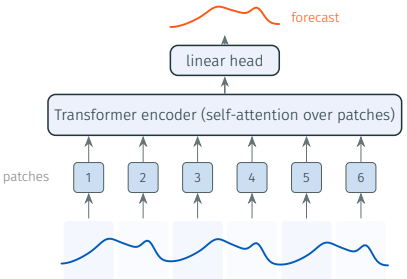
Other critiques

- Attention ignores order; one point means little
- Inconsistent setups, *drop-last* trick
- Tuned fairly: **no single champion**
- Weak baselines, evaluation pitfalls

[Brigato et al., 2025, Hewamalage et al., 2023, Qiu et al., 2024, Zeng et al., 2023]

The real question: what is a good **token** for a time series?

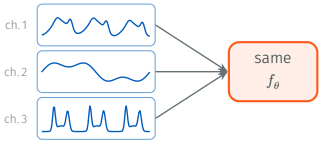
PatchTST (2023): the design behind most TSFMs



Key idea #1: patches as tokens

a token now carries a local shape, not a single value

Key idea #2: channel independence



+ instance normalization [Kim et al., 2022]

Shape- and scale-agnostic inputs: the first step towards transfer [Nie et al., 2023]

The lesson of a decade

Deep learning is relevant when...

- it can **pool many related series** in one global model
- the inductive bias fits (patches, convolutions)
- inputs are **normalized**

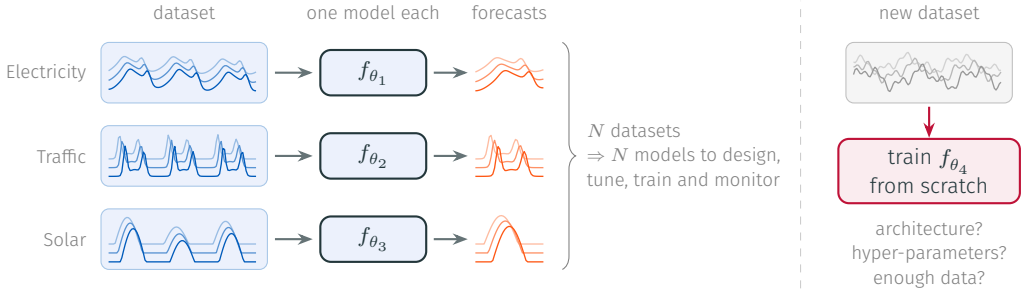
...and struggles when

- the dataset is **small and isolated**
- the distribution **shifts**
- the dataset is **new**: start again

Next step: pool series **across datasets**, not only within one

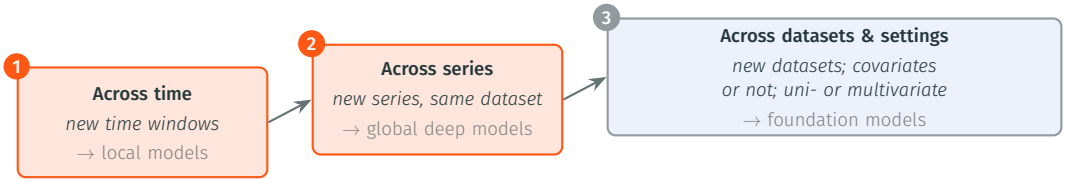
Part II: Limitations

Standard paradigm: one model per dataset, even for one task



Same task, new dataset $\Rightarrow$ new architecture, new tuning, new training: **start again**

Why this does not scale



- **Cold start**: a new sensor, customer or product has no history
- A **fleet of models** to train, tune and monitor
- Benchmarks only test levels 1–2 → say little about a **new** dataset

What we would like instead

Train **once**, on **many** series

$\Downarrow$

use on **new** series and datasets, with or without covariates

without retraining

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$\Downarrow$

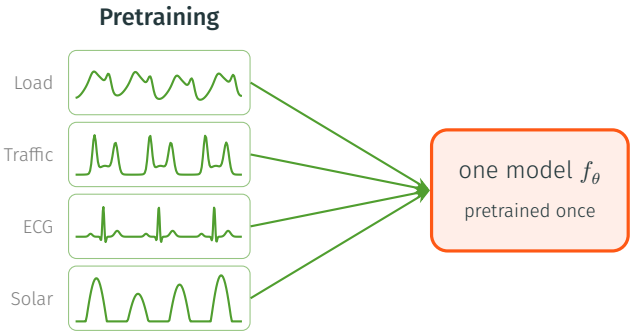
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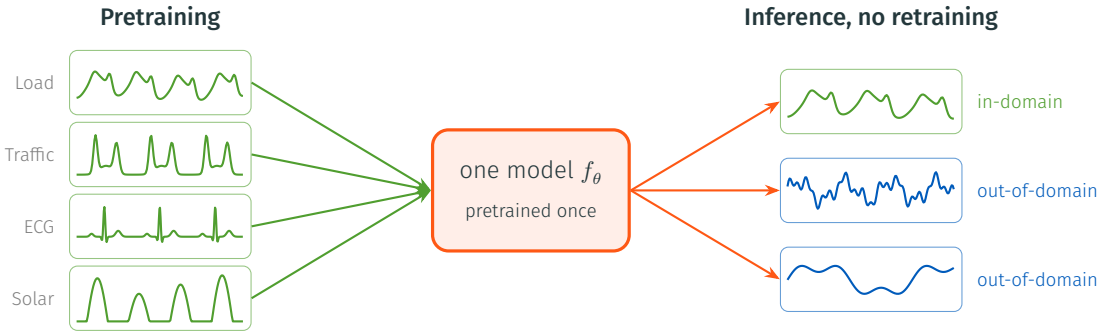
Ingredients: good **tokenization** + efficient **inference** + lots of **diverse data**

Part III: Foundation models

A shift with foundation models: train once, use everywhere

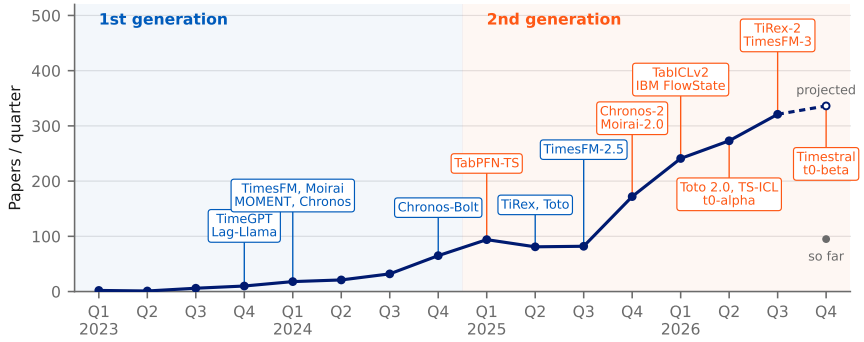


A shift with foundation models: train once, use everywhere



In-domain: new series of datasets seen in pretraining
Out-of-domain: entirely new datasets (zero-shot)

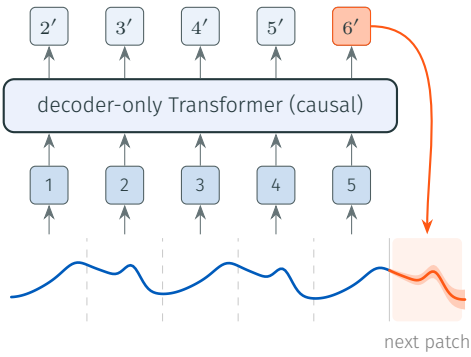
An explosion of time series foundation models (TSFMs)



Papers per quarter mentioning “time series” and “foundation model(s)”, OpenAlex (26 Oct. 2026). Q4 projected from its first 26 days. Labels: main releases.

An **increasingly present** topic: a few papers per quarter in 2023, more than 300 in 2026

1st generation (2023–2025): scaling patch-based Transformers



- TimesFM, Moirai, Toto…
- Pretrained on **billions** of time points, mostly real (some synthetic)
- Patches as tokens, a Transformer forecasts (or reconstructs) them

Compete with supervised baselines… but **univariate** (or multivariate), **no covariates**

2nd generation (2025–2026): adapt to the setting at inference

CHRONOS-2 · TIMESFM-3 · TIREX-2 · T0-GAMMA · TIMESTRAL · TS-ICL · ETC.

Covariates



Mix target, covariates and channels inside the model

CHRONOS-2, TIMESFM-3, TIREX-2, T0, TS-ICL, ETC.

Synthetic data



Not new, but much richer: **synthetic tasks with covariates**

CHRONOS-2, TIMESTRAL, CAUKER, ETC.

In-context learning

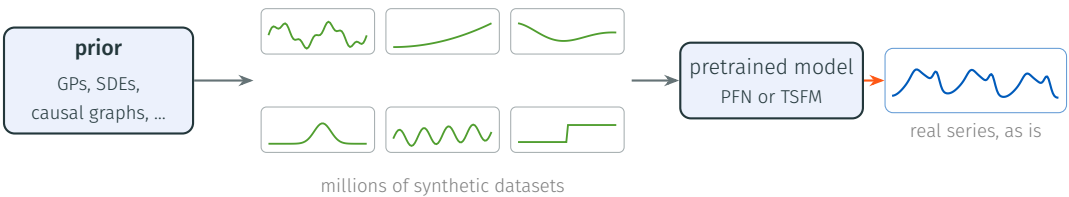


Adapt on the fly to examples given in the input

TABPFN-TS, TS-ICL, TIMEE (classification), ETC.

A great level of generalization: new datasets, with or without covariates, uni- or multivariate

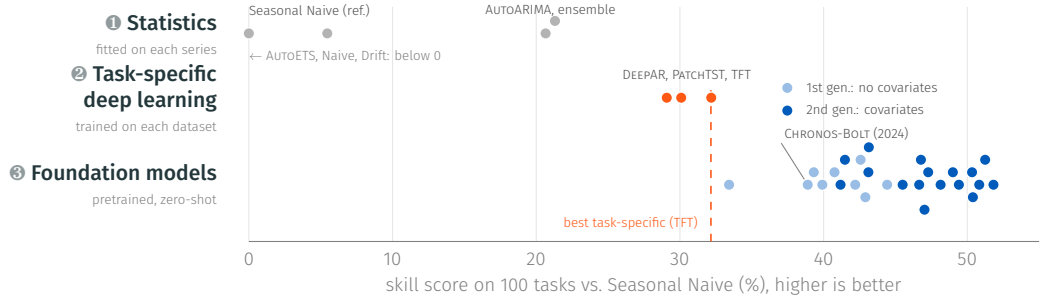
Key idea: pretrain on synthetic tasks sampled from a prior



A prior gives **diversity by design**: TempoPFN is pretrained on synthetic data only

Tabular PFNs truly learn from the context; most TSFMs only condition on the series' own history [Hollmann et al., 2025, Moroshan et al., 2026]

Learning from many datasets pays off: the gap on `fev-bench`



One dot per model shown on the leaderboard (2 Nov 2026) [Shchur et al., 2025], probabilistic skill score (SQL).

All 26 foundation models beat the best task-specific deep model

Today's programme: where each talk fits

Time	Talk	Link with this talk
11:00	CAUKER, TABBY etc. – S. Xie	**Representations and scalable** pretraining
13:30	TABICLV2 – G. Varoquaux	**In-context learning**, tabular FMs
14:10	TS-ICL – T. Nabil, E. Le Naour	**ICL for time series**: forecasting & imputation
15:15	Retrieval over training – R. Stanzione	**Anomaly detection**: which pretrained model to call at inference? An alternative to a TSFM
15:55	Decathlon – R. Nedellec	**In production**: TSFMs in operation

Conclusion

Take-home messages

Take-home messages

1. Ten years of deep learning for time series: **one model, one dataset, one task**
2. Tokenization, normalization and lots of data made **foundation models** possible
3. The new frontier: **flexibility at inference** (new datasets, covariates, multivariate)

Open questions for the day

1. Real or **synthetic** data?
2. Zero-shot, fine-tuning or **in-context learning**?
3. Do foundation models **memorize** or **generalize**?
4. Do **scaling laws** exist for TSFMs?

Thanks for listening!

[Backup] The landscape in October 2026

Model	Org.	Release	Params	What is new
TabPFN-TS	Prior Labs	Jan. 2025	–	Tabular FM + simple time features, covariates for free
TiRex	NXAI	May 2025	35M	xLSTM backbone, state tracking for long and short horizons
TimesFM-2.5	Google	Sep. 2025	200M	Smaller than v2.0, 16k context, quantile head
Chronos-2	Amazon	Oct. 2025	120M	Group attention: multivariate & covariates via ICL
Moirai 2.0	Salesforce	Nov. 2025	∼14M	Decoder-only, 30× smaller than Moirai-L and better
TabICLv2	Inria	Feb. 2026	–	Open, faster and scalable tabular FM
FlowState r1.1	IBM	Mar. 2026	∼19M	SSM encoder + functional decoder, sampling-rate invariant
Toto 2.0	Datadog	May 2026	4M–2.5B	A single recipe that scales (u-μP)
TS-ICL	EDF	Jun. 2026	28M	Time-indexed ICL: forecasting & imputation, covariates
TiRex-2	NXAI	Jul. 2026	38M (+44M)	Multivariate & covariates, constant-cost streaming
TimesFM-3	Google	Aug. 2026	330M	Native multivariate & covariates in one pass (non-commercial weights)
Timestral	Mistral AI	Oct. 2026	5M–209M	Mostly synthetic pretraining, 32k context, CPU inference

Almost every release claims the top of GIFT-Eval, fev-bench or TIME: leaders change **every few months**, with small gaps

[Backup] Beyond forecasting

Classification

- Multi-task FMs: MOMENT [Goswami et al., 2024]
- Dedicated encoders: MANTIS [Feofanov et al., 2025], MANTISV2 [Feofanov et al., 2026]
- Pretraining on **synthetic data only**: CAUKER [Xie et al., 2026]

Imputation

- *Time-indexed* FMs lead a 33-dataset zero-shot benchmark [Le Naour et al., 2025b]: TABPFN-TS, MOTM [Le Naour et al., 2025a]
- One model for forecasting *and* imputation: TS-ICL [Le Naour et al., 2026]

Anomaly detection

- Zero-shot scoring with FMs: MOMENT, tiny TSPULSE [Ekambaram et al., 2026]
- Still no single winner → **select** the right detector for a new series [Panourgias et al., 2026]

Generalization across **tasks** (level 3) is the open frontier

[Backup] How do we evaluate TSFMs?

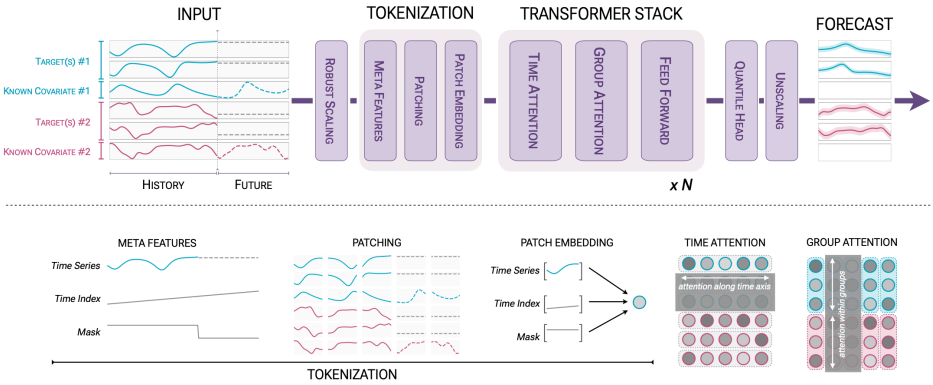
Zero-shot forecasting benchmarks

- GIFT-Eval: 97 configurations + a non-leaking pretraining corpus [Aksu et al., 2024]
- fev-bench: 100 tasks, 30 with covariates [Shchur et al., 2025]
- TIME: 50 new datasets, 98 tasks, built for strict zero-shot [Qiao et al., 2026]
- BOOM: observability data [Cohen et al., 2025]

Open issues

- **Leakage**: test series hidden in huge pretraining corpora
- Leaderboards mix zero-shot, fine-tuned and ensembles
- Small gaps between top models, metric-dependent rankings
- Few benchmarks beyond forecasting
- Benchmarks $\neq$ **operations**

[Backup] CHRONOS-2: in-context learning meets covariates



CHRONOS-2 architecture [Ansari et al., 2025].

Group attention shares information within a group (variates, target + covariates, related series); trained on **synthetic** multivariate structures → zero-shot covariate-aware forecasting

[Backup] Limitations of the 1st generation

Main limitations

1. Lack of rigorous unified benchmarks [Hewamalage et al., 2023]: GIFT-Eval (Dec. 2024) as a first effort [Aksu et al., 2024]
2. Struggle to clearly outperform the best supervised baselines, may require fine-tuning [Goswami et al., 2024]
3. **Univariate-only**: unable to use **covariates** or cross-channel information
4. **Forecasting-only**, on regular and complete inputs

[Backup] What makes time series hard for deep learning?

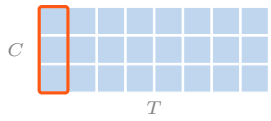
Five specificities

1. **Temporal structure**: order matters, short- and long-range dependencies, multiple periodicities
2. **Non-stationarity**: the distribution drifts over time → train/test shift is the norm
3. **Low semantic density**: one time step carries little information (unlike a word) → what is the right *token*?
4. **Heterogeneity**: number of channels, length, sampling rate and units vary across datasets
5. **Small data per dataset**: benchmarks are tiny compared to vision or NLP

Each point motivates an architectural choice… and, eventually, the move to **foundation models**

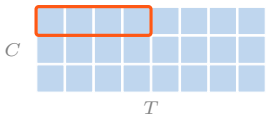
[Backup] What is a token?

Point (2017–2022)



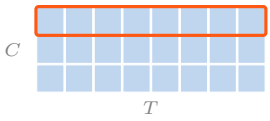
One token = all channels at time t. Little semantics, long sequences.

Patch (PatchTST)



One token = a sub-series of one channel. Local shape, **channel independence**.

Variate (iTransformer)



One token = a whole channel; attention models **cross-channel** dependencies [Liu et al., 2024].

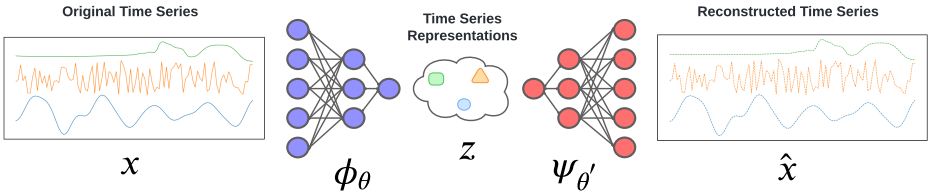
Patches are the tokenization adopted by most TSFMs; variate-level mixing comes back with covariates (2nd generation)

[Backup] The tricks that matter (often more than the backbone)

- **Instance normalization** (RevIN): normalize each input window, de-normalize the output $\rightarrow$ robustness to distribution shift [Kim et al., 2022, Liu et al., 2022]
- **Decomposition** trend / seasonality [Wu et al., 2021, Zeng et al., 2023]
- **Channel independence vs. dependence**: a capacity / robustness trade-off [Han et al., 2024]
- **Multi-periodicity**: reshape 1D series into 2D (period $\times$ phase), TimesNet as a **task-general** backbone [Wu et al., 2023]
- **State space models** (S4, Mamba): linear-time long-range modelling [Gu and Dao, 2023, Gu et al., 2022]

Normalization + patching + channel independence = **shape- and scale-agnostic** models, the ingredients of cross-dataset transfer

[Backup] Learning representations without labels



Self-supervised objectives

- **Contrastive**: T-Loss [Franceschi et al., 2019], TS2Vec [Yue et al., 2022]
- **Masked reconstruction**: TST [Zerveas et al., 2021], PatchTST [Nie et al., 2023]
- **Hybrid**: SimMTM [Dong et al., 2023]

Leading to

- Few-shot classification, reusable representations
- Transfer... but pretraining on the **same** dataset or domain

[Backup] From architectures to tasks (1/2): predicting values

Forecasting

- **Point** vs. **probabilistic** (quantiles, CRPS) [Lim et al., 2021, Salinas et al., 2020]
- **Short-term** (M4, Monash) [Godahewa et al., 2021, Makridakis et al., 2020] vs. **long-term** (ETT, Electricity, Traffic…)
- Exogenous covariates (calendar, weather) are crucial in practice [Lim et al., 2021]… but rare in academic benchmarks
- Best models: patch Transformers, MLP-mixers, linear models, with **small and unstable gaps** [Qiu et al., 2024, Wang et al., 2024]

Imputation

- Missing values are everywhere (sensor failures, transmission losses)
- **Deterministic**: BRITS [Cao et al., 2018], SAITS [Du et al., 2023]
- **Probabilistic**: CSDI, score-based diffusion [Tashiro et al., 2021]
- **Continuous-time**: implicit neural representations, irregular sampling, impute *and* forecast [Le Naour et al., 2024]
- Masked reconstruction = also a **pretraining objective**

[Backup] From architectures to tasks (2/2): labels and scores

Classification & regression

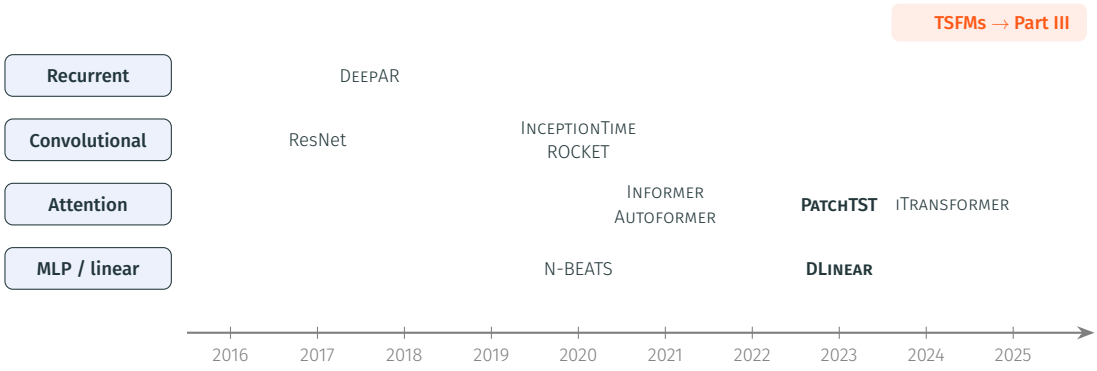
- Benchmarks: UCR (128 univariate), UEA (multivariate) [Bagnall et al., 2018, Dau et al., 2019], TSER for regression [Tan et al., 2021]
- ResNet, InceptionTime, ROCKET family [Dempster et al., 2020, Ismail Fawaz et al., 2019]
- Many datasets with **only tens to hundreds** of labelled series

Anomaly detection

- Mostly **unsupervised**: learn "normal", score deviations (reconstruction, prediction error, association discrepancy) [Malhotra et al., 2016, Xu et al., 2022]
- **Fragile evaluation**: trivial or mislabelled anomalies, point-adjusted F1 [Liu and Paparrizos, 2024, Wu and Keogh, 2023]
- No method wins everywhere $\rightarrow$ **model selection**

Small labelled sets and heterogeneous data $\Rightarrow$ strong appeal of **pretrained** models

[Backup] A decade at a glance



[Backup] Five tasks, five families of specialized models

Task	Typical models	Typical benchmarks
Forecasting	DeepAR, PatchTST, DLinear	M4, long-term (ETT, Traffic…)
Classification	InceptionTime, ROCKET	UCR / UEA archives
Extrinsic regression	same backbones, regression head	TSER archive
Imputation	BRITS, SAITS, CSDI	masked public datasets
Anomaly detection	reconstruction / prediction error	TSB-AD

Ten years of progress… but almost always **one model, one dataset, one task**

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